

Mohit Dubey

AI Full Stack Engineer

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WORK EXPERIENCE

Software Engineer — Publicis Sapient

June 2025 - Present

- Built a **schema-driven / server-driven UI (SDUI)** renderer powering dynamic layouts, conditional fields, async data bindings, and versioned JSON configs — enabling product teams to ship configuration-driven experiences without redeployments, cutting UI rollout time significantly.
- Designed a **reusable, highly configurable table component** adopted as a shared platform primitive across 15+ modules and multiple product teams, improving UI consistency and reducing per-team maintenance overhead.
- Led adoption of a **federated design system** across 50+ teams, establishing shared component patterns, guardrails, and contribution standards to raise the quality bar across product surfaces.
- Centralized cross-service communication via a **shared network/API layer** with consistent error handling and retry logic, improving reliability and developer velocity for teams building on top of the platform.
- Architected **micro-frontend** boundaries and frontend infrastructure using React.js and Next.js, improving independent deployability and platform performance by 60% across product teams.
- Built an internal **MCP server** with dynamic tool discovery and context orchestration for AI coding agents, reducing LLM context overhead by 70% — part of a broader push toward **AI-native developer experience** and agentic engineering workflows.
- Designed a **client-server UI version synchronization system** (React + Nginx) as shared platform infrastructure, ensuring consistent builds across all consuming product teams.
- Applied **code-splitting, lazy loading, and dynamic imports** as reusable performance guardrails, reducing JavaScript bundle size and improving time-to-interactive across the platform.
- Designed and implemented an **MCP server** with dynamic tool discovery, semantic search, and context orchestration, reducing LLM context overhead by 70% and lowering token consumption and inference costs across production workloads.
- Architected scalable agent workflows leveraging **LangGraph**, vector databases, reranking, and hybrid retrieval techniques, increasing answer relevance by 42% and supporting thousands of concurrent enterprise requests.
- Managed the full project delivery lifecycle from conception to production rollout, maintaining a consistent **95% project success rate**.
- Designed and implemented a **client-server UI version synchronization system** (React + Nginx) to ensure users always run the latest deployed build.
- Built a **reusable, highly configurable table component** adopted across 15+ modules, improving consistency and reducing maintenance overhead.
- Centralized API interactions by creating a **shared network layer** with consistent error handling and retry logic, improving reliability across the application.
- Implemented **performance-optimization** techniques such as **code-splitting, lazy loading, and dynamic imports** to reduce **JavaScript bundle size** and improve time to interactive (TTI).
- Engineered scalable **Frontend Architecture** using **React.js, Next.js**, and micro-frontends, optimizing platform performance by **60%** and securing workflows using **OAuth2** authentication protocols.

Software Engineer — Deqode

Aug 2023 - May 2025

- Built a **schema-to-React renderer** supporting dynamic layouts, conditional fields, async data bindings, and versioned JSON configuration — a configuration-driven UI framework consumed by multiple product teams.
- Owned frontend infrastructure modernization: migrated **Redux** to modular **RTK** slices + RTK Query, reducing API overfetching by 78% and standardizing state-management patterns across teams.
- Built a developer-facing **virtualized grid** component (1M+ rows) using **react-window**, published as a reusable platform primitive to reduce implementation cost for feature teams.
- Led a **federated design system** adopted by 50+ teams, enforcing accessibility, performance, and API consistency standards (ARIA roles, keyboard navigation, semantic HTML) across the component library.
- Instrumented **Web Vitals and Sentry** across 500K+ sessions/month to establish platform-wide performance guardrails (LCP, FCP, TTI), surfacing regressions before they reached product teams.
- Partnered with 15+ engineering teams to modernize deployment infrastructure (Kubernetes, MLflow, Terraform), compressing release cycles from 3 weeks to 2 days — direct experience owning platform tooling that multiple teams depend on.
- Orchestrated cloud-native ML systems on **Kubernetes** with **MLflow**, Terraform, and Vertex AI Pipelines; led platform modernization for **15+ ML teams**, compressing model deployment cycle from 3 weeks to 2 days.
- Created robust **AI monitoring and real-time dashboarding infrastructures** using Grafana, Power BI, and Looker to continuously track operational metrics, system health, and model drift.
- Built Aadhaar-Voter ID linking UI in React, scaling to **scaling to handle high-traffic production loads** with resilient, reusable components.
- Migrated **Redux** to modular **RTK** slices + RTK Query, reducing API overfetching by 78%.
- Developed editable virtualized grid (1M+ rows) using **react-window**, with smooth scroll and memoization.
- Led federated design system adopted by 50+ teams, enforced standards.

- Integrated Web Vitals + **Sentry** to track LCP, FCP, TTI across 500k+ sessions/month and optimize core web metrics.
- Implemented **keyboard navigation**, focus management, ARIA roles, and screen reader support using semantic HTML.
- Built schema-to-React renderer supporting **dynamic layouts**, conditional fields, async data bindings, and versioned JSON configs; cut UI rollout time.

Software Engineer — Shiv Technolabs

Feb 2023 - July 2023

- Leveraged deep knowledge of browser internals to optimize performance, systematically reducing reflow,repaint, and bundle sizes to achieve significant improvements in Core Web Vitals.
- Implemented advanced state management patterns across complex application workflows using MobX and modern ES6+ JavaScript features for predictable data flow.
- Led frontend performance audits analyzing JS engines compilation behaviors and execution contexts to eliminate memory leaks and runtime bottlenecks.
- Orchestrated the transition to **Next.js** methodologies, integrating **RESTful APIs** to achieve seamless data transfer across platforms, which decreased website bounce rate by **15%** and grew traffic.

Software Engineer — Sourcebae

May 2021 - Jan 2023

- Built full-stack applications with **React**, **TypeScript**, and **NestJS**, shipping REST API backends backed by **PostgreSQL** and **MongoDB** serving 100K+ users with sub-100ms p99 latency via Redis caching.
- Designed a reusable **TypeScript** component library with centralized state management, cutting feature delivery time by 40% across 5 concurrent product teams.
- Containerized services with **Docker** and established **CI/CD pipelines** with automated testing and staged rollouts, eliminating manual deployment errors and enabling zero-downtime releases.
- Built **offline-first UI** in React Native using SQLite for local data storage and background sync to update the server when reconnected.

PROJECTS

TradeBrains — Automated Real-Time Algorithmic Trading Platform

- Architected an end-to-end algorithmic trading platform predicting 20-minute stock movements by fine-tuning open-source generative models (**Llama 3.3 70B**) using 3-stage **RLHF (SFT, Reward Model, PPO)** and **GRPO** to align prediction outputs directly with target performance metrics.
- Implemented custom **GRPO/PPO reward functions** integrating multi-objective metrics (factual correctness, formatting compliance, and financial risk metrics) to eliminate hallucinations in high-stakes domain outputs.
- Engineered **QLoRA 4-bit** parameter-efficient fine-tuning (**PEFT**) pipelines for 8B-parameter sentiment models, reducing memory footprints while maintaining **88% structured score extraction accuracy**.
- Executed **Direct Preference Optimization (DPO)** and **KTO** fine-tuning on domain-specific instruction datasets, bypassing separate reward model latency while matching RLHF alignment quality.

EDUCATION

Rajiv Gandhi Proudhyogiki Vishwavidyalaya

Bhopal, India

Bachelor of Engineering (HONOURS) - Information Technology; GPA: 7.75

August 2016 - May 2020

ACHIEVEMENTS

- [AWS Certified Developer Associate](#).
- [AWS Certified AI Practitioner](#).
- [AWS Certified Cloud Practitioner](#).
- [Hacker Rank Javascript Basic](#).
- [Hacker Rank CSS certification](#).
- [Sololearn React + Redux](#)

TECHNICAL SKILLS

- **LLMs & Agents:** LangChain, LangGraph, CrewAI, Google ADK, LlamaIndex, Hugging Face Transformers, OpenAI API, Ollama, vLLM, TensorRT-LLM, MCP Servers, Agent Orchestration, RAG Systems, Prompt Engineering
- **Training & Optimization:** LLM Fine-Tuning (SFT, QLoRA, LoRA, PEFT), RLHF, GRPO, DeepSpeed, FSDP, Distributed Training, CUDA, Model Quantization (INT8, NF4)
- **ML & Deep Learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, Classical ML, Feature Engineering, Model Optimization
- **Data Engineering:** Apache Spark (PySpark, Spark SQL), Databricks, Delta Lake, Kafka, Airflow, BigQuery, ETL/ELT, Feature Store (Feast)
- **MLOps & Infrastructure:** MLflow, Kubeflow, Kubernetes, Docker, Terraform, CI/CD Pipelines, Model Monitoring, Drift Detection, A/B Testing
- **Cloud:** AWS (SageMaker, EKS, Lambda, S3), GCP (Vertex AI, GKE, Dataproc, BigQuery)
- **Backend Engineering:** Python, FastAPI, Go, TypeScript, REST APIs, gRPC, Microservices, WebSocket, Redis, n8n
- **Databases & Vector Stores:** PostgreSQL, MongoDB, Redis, Qdrant, Milvus, Weaviate, Chroma, SQL
- **Frameworks:** Reactjs, Redux, Nextjs, SCSS, MaterialUI, ExpressJs, Jest, RTL, Enzyme
- **Tools:** Azure, Docker, Webpack, Git/Github, Postman, Swagger, Figma, Sequelize ORM, Firebase

- **Coursework:** System Design, Database Management Systems, Data Structures and Algorithms, Object Oriented Programming
- **Other:** RESTful Web Services, Agile methodologies, Integration testing, micro frontends